Ryan Peckham

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Education

University of Michigan, Seeking **Computer Engineering and Applied Physics,** Ann Arbor, MI — Aug. 2025 – Apr. 2029
Coursework*: Multivariable & Vector Calculus, Electricity & Magnetism, Quantum Photonics, Discrete Mathematics, Solar Energy Systems, Mandarin, Probabilistic Statistics* — GPA: 3.35 / 4.0

Skills

Programming languages: Python, C, C++, C#, Rust, Java, SQL, Front-End Stack, MATLAB
Systems & Concepts: Git & GitHub, PostgreSQL, WSL, Docker, Database Normalization, Threading
Operating Systems: Windows, Linux (Ubuntu, Debian, NixOS), MacOS
Hardware: Soldering, Shielded Metal Arc Welding, CNC Machines, Lathes, Mills

Experience

UNIVERSITY OF MICHIGAN

Michigan Autonomous Security Team (M-AST), Chief Software Engineer — Jan. 2026 - Present

- Leading and instructing a team of engineers to install, program, and test gyroscope, magnetometer, barometer, and GPS sensors onto a prototype drone
- Maintaining the organization of a large and complex codebase and reviewing code and pull requests
- Collaborated with M-AST President and peers to design the drone architecture specifying the flow of information between systems, including the implementation of SITL and HITL runtimes
- Integrated several software architectures including ROS2, OpenCV, telemetry code, and path-finding

Michigan Hackers, Machine Learning Team, Member — Nov. 2025 - Present

- Learned and applied fundamental computer science concepts related to machine learning such as Markov Chains, Bayes' Theorem, and the various kinds and applications of machine learning
- Attended regular lectures by other students while discussing and practicing new material with peers

Course: Engineering 100 — Aug. 2025 – Dec. 2025

As part of a 5-member engineering team, I completed the C++ programming, wiring, and soldering of a sun-tracking solar panel using photoresistors, stepper motors, voltage limiters, and a battery on a RaspberryPi platform. We engaged in a comprehensive engineering process including finding a problem in modern technology, thoroughly researching the subject, and then presenting a proposed solution to Engineering faculty. Following the project, we tested our solution and presented it, data, and conclusions to faculty. This project was practice for completing engineering projects in professional situations and taught me communication, collaboration, task prioritization, how to conclude meaningful results from data, and how to balance planning with execution.

PERSONAL PROJECTS

Designed and developed my personal website https://peckhamr.com using HTML, JS, CSS, and the Svelte UI framework. Used as an opportunity to teach myself the foundations of domain registrars, hosting a website, and programming a maintainable codebase. In addition, most of what I know about computer science is material which I first taught myself in small personal projects, such as simple interpreted languages, Sudoku solvers, and API wrappers.

WEST OTTAWA HIGH SCHOOL

FIRST Robotics Competition (FRC), Team 141 "WOBOT", Programming Sub-team — Aug. 2023 – Oct. 2024

- Developed input and movement code by implementing and tuning PIDF loops for precise control
- District Champions; Awards won by team: Engineering Inspiration, Sustainability, Entrepreneurship

Cultural, Technological, and Environmental Exchange (CTEE) Club — Sep. 2024 – Apr. 2025

- Repaired and tested 2-3 community-donated desktop computers monitors each week
- Learned about the architecture and design of desktop computers & motherboards
- Replaced faulty components safely and prepared ready-to-use operating systems (Windows)
- Stored repaired units for future shipment to developing nations and communities